Q1. Define **multisensor navigation** and explain why more than one navigation sensor is normally required.

Answer:

Multisensor navigation is the process of estimating the aircraft's navigation state—such as **position, velocity, and attitude**—by combining measurements from two or more navigation sensors.

A single sensor cannot normally provide all required navigation variables with sufficient **accuracy, reliability, and continuity**.

For example:

- INS provides continuous position, velocity, and attitude, but its errors grow with time.
- GPS provides accurate position and velocity, but depends on external satellite signals.
- Doppler radar provides ground velocity.
- A star tracker provides an external attitude reference.

Therefore, multisensor navigation combines **complementary sensors** so that the weaknesses of one sensor can be compensated for by another.

Multisensor navigation = combination of complementary sensors

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Q2. Differentiate between:

a) Dead-reckoning sensors
b) Positioning sensors

Give **two examples** of each.

Answer:

1. Dead-reckoning sensors

These provide information related to **acceleration or velocity**, which must be integrated with time to obtain position.

Examples:

- INS
- Doppler radar

- Air-data sensors

2. Positioning sensors

These provide measurements related directly to the aircraft's **position relative to Earth**.

Examples:

- GPS
- Loran
- Star tracker in suitable navigation configurations

Main difference:

Dead reckoning → integrate measurements

while:

Positioning sensor → external position reference

Q3. Explain the basic error propagation mechanism in an INS from:

sensor error → velocity error → position error

Answer:

The INS uses accelerometers to measure specific force and gyroscopes to measure angular motion.

An error in the accelerometer measurement produces an error in the calculated acceleration.

Integrating acceleration once gives velocity:

$$\Delta V = \int \Delta a \, dt$$

and integrating velocity gives position:

$$\Delta P = \int \Delta V \, dt$$

Therefore: Accelerometer error → Velocity error → Position error

Gyro errors can also produce attitude errors. An attitude error changes the transformation of measured acceleration into the navigation frame, which can create additional velocity and position errors.

Thus INS errors can propagate through several interconnected states.

Q4. Why is a **star tracker** a useful complementary sensor for an INS?

Answer:

A star tracker provides an **external reference for aircraft/platform orientation** by observing known stars.

One important INS error source is **gyro drift**. Gyro drift causes the calculated orientation of the INS platform to gradually deviate from its true orientation.

The star tracker provides an independent attitude reference that can be compared with the INS attitude.

Therefore, the difference can be used to estimate gyro-related errors.

Star tracker → external attitude reference → gyro-drift estimation

This helps prevent the long-term divergence of the INS solution.

Q5. Explain how Doppler radar can help limit the growth of INS navigation errors.

Answer:

Doppler radar provides an external measurement of the aircraft's **velocity relative to the ground**.

The INS independently calculates velocity.

Therefore, the two measurements can be compared:

$$\Delta V = V_{INS} - V_{Doppler}$$

The resulting velocity difference provides information about INS errors.

A Kalman filter can use this information to estimate errors in:

- velocity,
- position,

- platform alignment,
- sensor errors.

Therefore: Doppler velocity reference → INS error estimation/correction

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Q6. What is meant by **position updating** of an inertial navigation system?

Mention two possible external sources that can provide position updates.

Answer:

Position updating is the process of using an external position measurement to correct the position calculated by the INS.

For example: P_{INS}
can be compared with: P_{GPS}

to obtain: $\Delta P = P_{INS} - P_{GPS}$
The external position information can come from systems such as:

- GPS
- Loran
- other radio navigation systems

A Kalman filter can use the position difference not only to estimate position error, but also correlated errors such as:

- velocity errors,
- attitude errors,
- sensor errors.

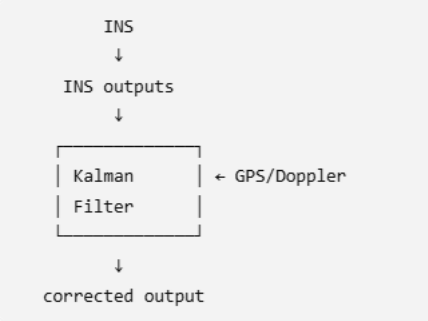
Q7. Explain the difference between: Open-loop Kalman filter and
Closed-loop Kalman filter.

Answer:

Open-loop

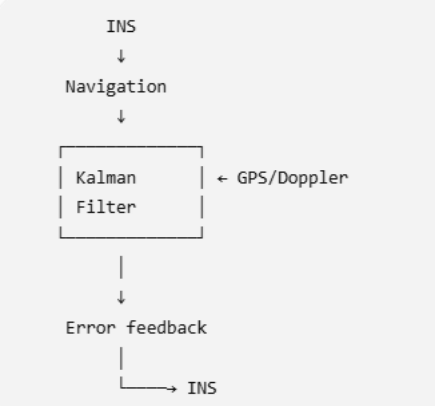
In an **open-loop** system, the Kalman filter estimates the INS errors and applies the corrections to the **INS outputs**.

The corrections are not fed back into the internal INS mechanization.



Closed-loop

In a **closed-loop** system, estimated errors are fed back into the INS to correct its internal navigation solution.



Main difference:

Open loop: correction to output

Closed loop: correction fed back to INS

Q8. Covariance Matrix

For the covariance matrix

$$P = \begin{bmatrix} 4 & 1.5 \\ 1.5 & 9 \end{bmatrix}$$

answer the following:

a) What is the variance of state 1?

b) What is the variance of state 2?

c) What is the covariance between the two states?

d) What does the off-diagonal term indicate physically?

a) Variance of state 1

Diagonal element: $\boxed{P_{11} = 4}$

b) Variance of state 2 $\boxed{P_{22} = 9}$

c) Covariance $\boxed{P_{12} = P_{21} = 1.5}$

d) Physical meaning

The diagonal elements represent the **uncertainty/variance** of each state.

The off-diagonal elements represent **correlation between the state errors**.

Therefore: $\boxed{P_{12} = 1.5}$
means that the two state errors are statistically correlated.

This correlation is particularly important in multisensor navigation because a measurement of one state can provide information about another correlated state.

Q9. Measurement Residual

An INS predicts a DME range of: $\rho^* = 10500\ m$
while the DME measurement is: $\rho = 10480\ m$
Calculate: $\Delta\rho$
Explain what the sign of the result means.

The residual is: $\Delta\rho = \rho^* - \rho$
Therefore: $\Delta\rho = 10500 - 10480$

$$\boxed{\Delta\rho = 20\ m}$$

Interpretation:

The predicted range is **20 m greater** than the measured range.

So the prediction and measurement disagree by: 20 m

Q10. Kalman Gain

For a scalar measurement, assume $P^- = 9$ and $R = 3$ with $H = 1$
Calculate the Kalman gain using: $K = P^- H^T (HP^- H^T + R)^{-1}$
Explain whether the filter gives more weight to the prediction or to the measurement.

The Kalman gain is:

$$K = P^{-}H^{T}(HP^{-}H^{T} + R)^{-1}$$

Substitute: $K = 9(1)[(1)(9)(1) + 3]^{-1}$

$$K = \frac{9}{12}$$

Therefore: $\boxed{K = 0.75}$

the filter gives relatively high weight to the **measurement** compared with the prediction.

Q11. Prediction and Update

A one-dimensional Kalman filter has:

$$\hat{x}^{-} = 10$$
$$P^{-} = 4$$

The measurement is: $z = 12$ with $H = 1,\ R = 2$, Calculate:

a) Kalman gain K

b) Measurement residual $r = z - H\hat{x}^{-}$
c) Updated state estimate $\hat{x} = \hat{x}^{-} + Kr$
d) Updated covariance $P = (1 - KH)P^{-}$

a) Kalman gain $K = \frac{P^{-}H}{HP^{-}H+R}$

$$K = \frac{4}{4 + 2}$$
$$\boxed{K = \frac{2}{3} \approx 0.667}$$

b) Measurement residual $r = z - H\hat{x}^{-}$

$$r = 12 - (1)(10)$$
$$\boxed{r = 2}$$

c) Updated state estimate $\hat{x} = \hat{x}^{-} + Kr$

$$\hat{x} = 10 + (0.667)(2)$$
$$\boxed{\hat{x} \approx 11.33}$$

d) Updated covariance $P = (1 - KH)P^{-}$

$$P = (1 - 0.667)(4)$$
$$\boxed{P \approx 1.33}$$

Notice that the updated estimate: 11.33 , lies between the prediction: 10 , and measurement: 12 which is exactly what we expect.

Q12. Nonlinear Range Measurement

A DME station is located at: $(x_1, y_1) = (0, 0)$ and the aircraft is located at:

$$(x, y) = (6000, 8000)\ m$$

calculate the aircraft-to-DME range.

The range is:

$$\rho = \sqrt{(x - x_1)^2 + (y - y_1)^2}$$

$$\rho = \sqrt{6000^2 + 8000^2}$$
$$\rho = \sqrt{36{,}000{,}000 + 64{,}000{,}000}$$
$$\rho = \sqrt{100{,}000{,}000}$$

$\boxed{\rho = 10{,}000\ m}$ $\boxed{\rho = 10\ km}$

Q13. For a range measurement , suppose: $\Delta x = 20\ m$, $\Delta y = -10\ m$ and $\theta = 60°$, Calculate the change in range. . Use answer of Q12 to calculate the new aircraft-to-DME range after the displacement.

$$\Delta\rho = \Delta x \cos\theta + \Delta y \sin\theta$$

$$\Delta x = 20\ m$$
$$\Delta y = -10\ m$$
$$\theta = 60°$$

$$\cos 60° = 0.5$$
$$\sin 60° = 0.866$$

$$\Delta\rho = (20)(0.5) + (-10)(0.866)$$

$\Delta\rho = 10 - 8.66$ = 1.34m

ρnew=ρ+Δρ = 1.34 + 10000=

Q14. Interpretation of Direction Cosines

explain why:

a) When $\theta = 0°$, the measurement is sensitive only to Δx.

b) When $\theta = 90°$, the measurement is sensitive only to Δy.

c) Why are $\cos\theta$ and $\sin\theta$ called direction cosines?

Equation: $\boxed{\Delta\rho = \Delta x \cos\theta + \Delta y \sin\theta}$

represents the projection of the aircraft position-error vector onto the **line of sight to the DME station**.

a) When $\theta = 0°$

$$\cos 0° = 1$$
$$\sin 0° = 0$$

$$\Delta\rho = \Delta x$$

So the range measurement is sensitive only to the x-direction error.

b) When $\theta = 90°$

$$\cos 90° = 0$$
$$\sin 90° = 1$$

$$\Delta\rho = \Delta y$$

So the measurement is sensitive only to the y-direction error.

c) Why direction cosines?

Because: $\cos\theta$ and $\sin\theta$

describe the components of the line-of-sight unit vector along the x- and y-axes.

Thus they determine how much of each position error contributes to the measured range error.

Q15. Explain the basic concept of **GPS–INS integration**.

Why is the combination generally more useful than using GPS or INS alone?

Answer:

GPS–INS integration combines:

INS Provides:

- continuous position
- velocity

- attitude
- high-rate navigation information

But its errors accumulate with time.

GPS Provides:

- external position information
- velocity information
- long-term reference

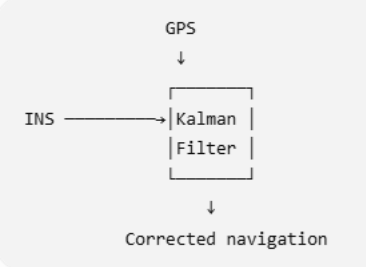
But GPS measurements are not necessarily continuous and depend on satellite signals.

Therefore, the two systems are complementary:

$$\boxed{INS = \text{continuous/high-rate}}$$
$$\boxed{GPS = \text{long-term external reference}}$$

A Kalman filter combines the information and estimates INS error states.

Conceptually:



Q16. Explain the role of the Kalman filter in a GPS–INS system.

In particular, explain why the filter does not simply "average" GPS and INS positions.

Answer:

Because the Kalman filter does much more than averaging two position values.

It estimates a **state/error vector** based on:

- INS dynamics
- sensor measurements
- process uncertainty

- measurement uncertainty
- correlations between error states.

For example, a GPS position residual may contain information about:

- position error
- velocity error
- attitude error
- gyro error
- accelerometer error
- clock errors.

Therefore, the filter uses the measurement residual to estimate the underlying error states.

$$\boxed{\text{GPS measurement} + \text{INS model} \rightarrow \text{error-state estimation}}$$

rather than:

$$\boxed{\frac{\text{GPS} + \text{INS}}{2}}$$

Q17.

What are the two GPS receiver clock states included in the simplified GPS–INS error model?

Explain the physical meaning of each.

The simplified GPS–INS model includes two GPS receiver clock states:

1. Clock bias

This represents the offset between the receiver clock and the reference GPS time.

is essentially a **time offset**.

It produces a corresponding error in measured pseudorange.

2. Clock drift

This represents the **rate of change of the clock bias**.

If the clock bias changes with time:

$$\frac{d(\text{clock bias})}{dt} = \text{clock drift}$$

Therefore: $\boxed{\text{Clock bias} = \text{clock offset}}$

$$\boxed{\text{Clock drift} = \text{rate of change of clock offset}}$$

These two states are included because receiver-clock errors affect GPS pseudorange measurements.

Q18.

List the major groups of states included in the **11-state GPS–INS error model**.

Organize your answer into:

- Position errors
- Velocity errors
- Platform misalignment errors
- GPS clock states

The simplified 11-state model can be grouped as follows.

1. Position errors — 3 states

$$\boxed{\Delta x,\ \Delta y,\ \Delta z}$$

These represent errors in the aircraft position.

2. Velocity errors — 3 states

$$\boxed{\Delta \dot{x},\ \Delta \dot{y},\ \Delta \dot{z}}$$

These represent errors in the calculated velocity.

3. Platform misalignment — 3 states

$$\boxed{\phi_x,\ \phi_y,\ \phi_z}$$

These represent orientation/misalignment errors of the inertial platform.

4. GPS clock states — 2 states

$$\boxed{\delta t,\ \delta \dot{t}}$$

where:

- δt= clock bias
- $\delta \dot{t}$= clock drift

Therefore: $3 + 3 + 3 + 2 = 11$

Q19. Compare open-loop and closed-loop Kalman filter mechanizations with respect to:

Item **Open Loop Closed Loop**

- Correction location
- Feedback to INS
- INS internal navigation solution
- Long-duration operation
- Implementation

Item	Open Loop	Closed Loop
Correction location	INS output	Inside/through feedback to INS
Feedback to INS	No	Yes
INS internal solution	Uncorrected/free INS solution	Corrected solution
Long-duration operation	Can be less convenient	Generally more robust
Implementation	Simpler conceptually	More integrated
Error estimates	Applied externally	Fed back to system

Key distinction:

$$\boxed{\text{Open loop: estimate} \rightarrow \text{correct output}}$$

$$\boxed{\text{Closed loop: estimate} \rightarrow \text{feed back} \rightarrow \text{correct INS}}$$

Q20. INS + Doppler Integration

An aircraft uses an INS and Doppler radar.

a) Explain the main error sources of the INS.

b) Explain what independent information the Doppler radar provides.

c) Explain how the Doppler measurement can help reduce INS error growth.

d) Explain why the combination is more robust than a standalone INS.

a) Main INS error sources

The two fundamental sensor error sources are:

Accelerometer errors and Gyro errors

Accelerometer errors affect velocity and position.

Gyro errors affect attitude, which can then introduce additional acceleration/velocity/position errors.

b) What does Doppler provide?

Doppler radar provides an external measurement of the aircraft's **velocity relative to the ground**.

Therefore: $V_{Doppler} \approx V_{ground}$

c) How does Doppler reduce INS error growth?

The INS velocity can be compared with the Doppler velocity:

$$\Delta V = V_{INS} - V_{Doppler}$$

This difference provides information about the INS error.

A Kalman filter can process this residual and estimate:

- velocity errors
- position errors
- platform alignment errors
- sensor errors.

The estimated errors can then be used to correct the navigation solution.

d) Why is the combination better than standalone INS?

INS provides: continuous, high-rate navigation

Doppler provides: external velocity reference

Therefore, the Doppler system can help prevent the INS errors from drifting freely.

Q21. INS + Star Tracker

An aircraft is equipped with an INS and a star tracker.

a) Explain why gyro drift is a major concern in an INS.

b) Explain how the star tracker provides an external reference.

a) Why is gyro drift a problem?

Gyroscopes measure angular motion.

A gyro drift error causes the calculated platform orientation to gradually deviate from its true orientation.

Since the INS uses its attitude to resolve accelerometer measurements into the navigation frame, attitude errors can create additional navigation errors.

b) How does the star tracker provide an external reference?

The star tracker observes known stars whose directions can be predicted accurately.

The measured star direction can be compared with the direction predicted by the INS.

Therefore, the difference provides information about INS orientation error.

Q22.

Two navigation systems use the same sensors.

System A uses a centralized Kalman filter, while System B uses several local filters followed by a master filter.

Explain:

a) The basic idea of the federated architecture.

b) Why it can be useful practically.

c) One disadvantage compared with a centralized implementation.

Solution

a) Basic idea of the federated architecture

A **federated Kalman filter architecture** divides the navigation estimation problem into several **local Kalman filters**. Each local filter processes information from a subset of the available sensors or navigation systems. Their estimated states or information are then sent to a **master filter**, which combines them to produce the final navigation solution.

In simple form:

Sensors → Local Kalman Filters → Master Filter → Final Navigation Solution

Unlike a centralized filter, where all sensor measurements are processed by one large Kalman filter, the federated approach distributes the processing among several filters.

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b) Why can it be useful practically?

The federated architecture has several practical advantages:

1. **Fault tolerance:** If one sensor or local filter fails, the remaining filters can continue providing navigation information.
2. **Modularity:** Different navigation sensors, such as INS, GPS, DME, or radar, can have their own local filters.
3. **Reduced computational complexity:** Each local filter handles a smaller estimation problem instead of one very large filter.
4. **Easy system expansion:** Additional sensors can be incorporated by adding another local filter without completely redesigning the main filter.
5. **Improved reliability:** Independent local estimates can be compared and monitored for sensor faults.

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c) One disadvantage compared with centralized implementation

The main disadvantage is that a **federated architecture can be less statistically optimal** than a properly designed centralized Kalman filter.

Because the information is divided among local filters, the master filter must combine their estimates carefully to avoid **double-counting correlated information**. This can make the

system more complex to tune and may result in slightly lower estimation accuracy than an ideal centralized implementation.

Short exam answer

a) A federated architecture uses several local Kalman filters, each processing a subset of sensor measurements, and a master filter combines their outputs to obtain the final navigation solution.

b) It provides modularity, fault tolerance, reduced computational requirements, and easier integration of different navigation sensors.

c) Its main disadvantage is that it may be less optimal than a centralized filter because correlations between information from different local filters can be difficult to handle and may lead to information double-counting.